

Biodegradation of Polyurethane Foam, Revisited, in the Rat Model

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In a prospective, randomized, controlled animal study, we systematically analyzed implanted polyurethane foam to determine the rate of degradation in the rat and to compare it to our previous human data. Sixteen 1-cm² silicone-backed pieces of polyurethane foam were randomly implanted into dorsal subcutaneous pockets in each of 16 Sprague-Dawley rats. Eight animals had polyurethane implants removed at 3 and 6 months and the remaining 8 animals at 9 and 12 months. Specimens were examined either by histology or by scanning electron microscopy of recovered foam after collagenase digestion of the capsule.

Histologically, there was evidence of foam degradation and a multinucleated giant cell inflammatory response surrounding the implants. With the scanning electron microscope, the strut width of the polyurethane foam upon implantation measured $51.4 \pm 1.3 \mu\text{m}$ (mean + SEM). This progressively decreased to $29.1 \pm 1.3 \mu\text{m}$ at 3 months, $16.6 \pm 0.6 \mu\text{m}$ at 6 months, $14.9 \pm 0.5 \mu\text{m}$ at 9 months, and $13.2 \pm 0.3 \mu\text{m}$ at 12 months ($p < 0.0001$).

Duration of implantation has a significant impact on polyurethane degradation as measured by scanning electron microscopy in the rat animal model. The rate of degradation in the rat is much faster than in our human study, indicating that the rat studies cannot be used to draw conclusions regarding rate of biodegradation in humans. (*Plast. Reconstr. Surg.* 96: 1326, 1995.)

Although the manufacturing of the polyurethane foam-covered breast prosthesis was discontinued in 1991, there is still a significant population of patients with these prostheses in place, and many unanswered questions remain concerning its biocompatibility and safety as an implantable device. Since it is impossible to obtain a well-controlled, prospective study of human polyurethane-covered breast implants, we have turned to an animal model to reanalyze polyurethane degradation. In designing a randomized, prospective, controlled animal study,

with the rat as the animal model, our objectives were (1) to analyze the effects that implantation time has on polyurethane degradation and (2), more importantly, to establish whether the rat is an adequate animal model to mimic the biodegradation of polyurethane documented in previous human studies.¹⁻⁴

MATERIALS AND METHODS

Medical-grade silicone elastomer sheets, polyurethane foam sheets, silicone glue, and washing solution were supplied by Surgitek, the manufacturer of the polyurethane-covered breast prosthesis. Following the company's protocol, which remains proprietary information, the materials were washed meticulously, and then a large polyurethane foam sheet was glued to a silicone elastomer sheet in order to mimic the structure of a silicone-backed polyurethane foam breast prosthesis. A second polyurethane foam sheet was excessively manipulated and manually stretched before being glued to another silicone elastomer sheet. This was designed to mimic rough intraoperative handling of an implant. Thus two types of silicone-backed foam sheets were formed: (1) using unmanipulated foam and (2) using stretched foam.

One hundred and twenty-eight 1-cm² silicone-backed polyurethane foam implants were then cut from each of the two prepared sheets. These implants were again treated to the washing protocol. The remaining portions of the elastomer-backed foam sheets were used as control foam samples.

Sixteen white Sprague-Dawley female rats weighing between 250 and 300 gm were anes-

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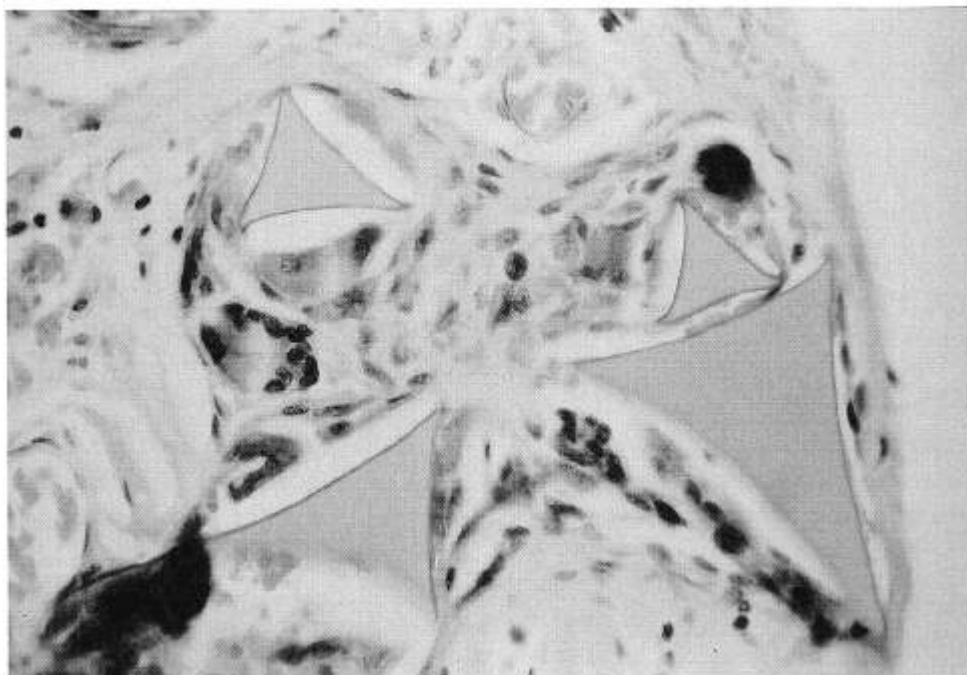


FIG. 1. Typical triangular shapes of the silicone-backed polyurethane foam implants in cross section. Note the multinucleated giant cell foreign-body response in the capsule around the foam after 3 months' implantation (Giemsa stain; $\times 400$).

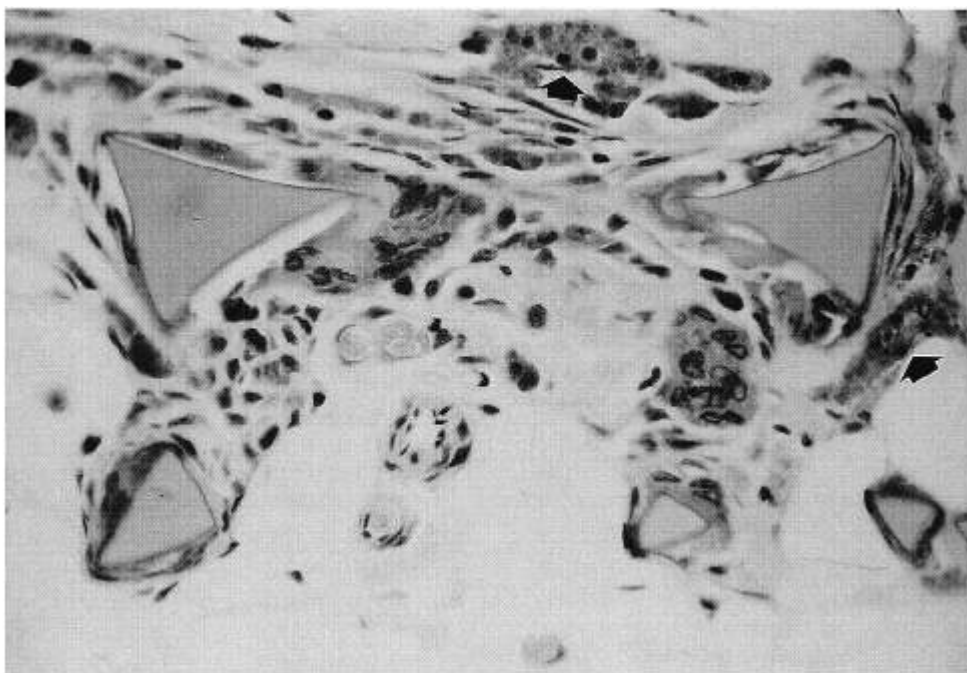


FIG. 2. Intracellular degraded foam particles (*arrow*) within many multinucleated giant cells after 9 months' implantation (Giemsa; $\times 400$).

thetized with 50 mg/kg of intraperitoneal pentobarbital. After shaving the dorsum of each animal and preparing with providine solution, eight unstretched and eight stretched silicone-backed foam implants were randomly inserted into each animal in separate subcutaneous

pockets that were then stapled closed. A pictorial map was drawn for each animal in order to record the placement of each implant. Postoperatively, the animals were allowed food and water ad libitum and given standard housing and care.

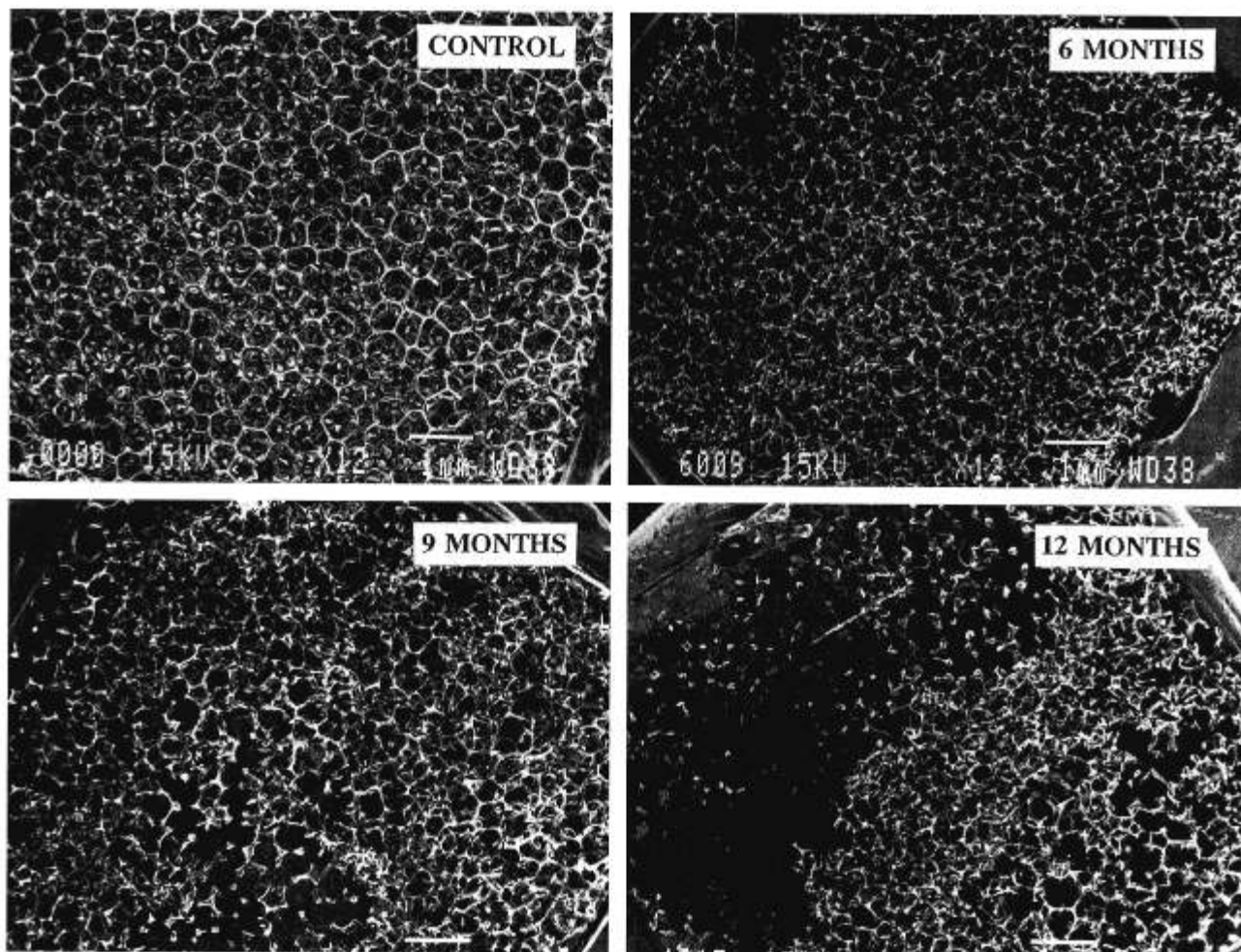


FIG. 3. Scanning electron micrographs showing the gradual loss of foam from the silicone elastomer backing as the duration of implantation increases ($\times 12$).

Eight animals were randomly chosen to be reanesthetized 3 months after implantation. The animals were weighed and anesthetized by the same protocol as outlined above. A group of four unstretched and a group of four stretched foam implants were randomly chosen from each animal and removed along with their respective tissue capsules. The implants in each group were then randomly assigned a number from 1 to 4. This assigned number would later determine the type of analysis the implant would undergo (see *Analysis*). The wounds were closed, and the same postoperative care described above was given to the animals. The implants assigned the number 1, with their surrounding tissue capsule, were fixed in a buffered 10% formalin solution as a preparation for histologic analysis. The other implants that were removed and numbered either 2, 3, or 4 were placed in a 0.005% w/v collagenase solution in order to digest the tissue capsule and to

retrieve the underlying foam. The tissue digestion process is thoroughly described in a previous publication.⁴ Each implant was then prepared for a specific analysis depending on the assigned number.

This same group of eight animals were reanesthetized 6 months after the initial implantation, and the remaining eight foam implants were removed. Exactly the same implant removal and preparation protocol was followed as described above except that the animals were then sacrificed with a lethal intraperitoneal injection of pentobarbital.

The remaining eight animals were left undisturbed until 9 months after implantation, when four stretched and four unstretched silicone-backed foam implants and surrounding tissue capsules were randomly chosen, removed, and prepared for analysis as described above. Twelve months after implantation, this same group of animals were reanesthetized, the re-

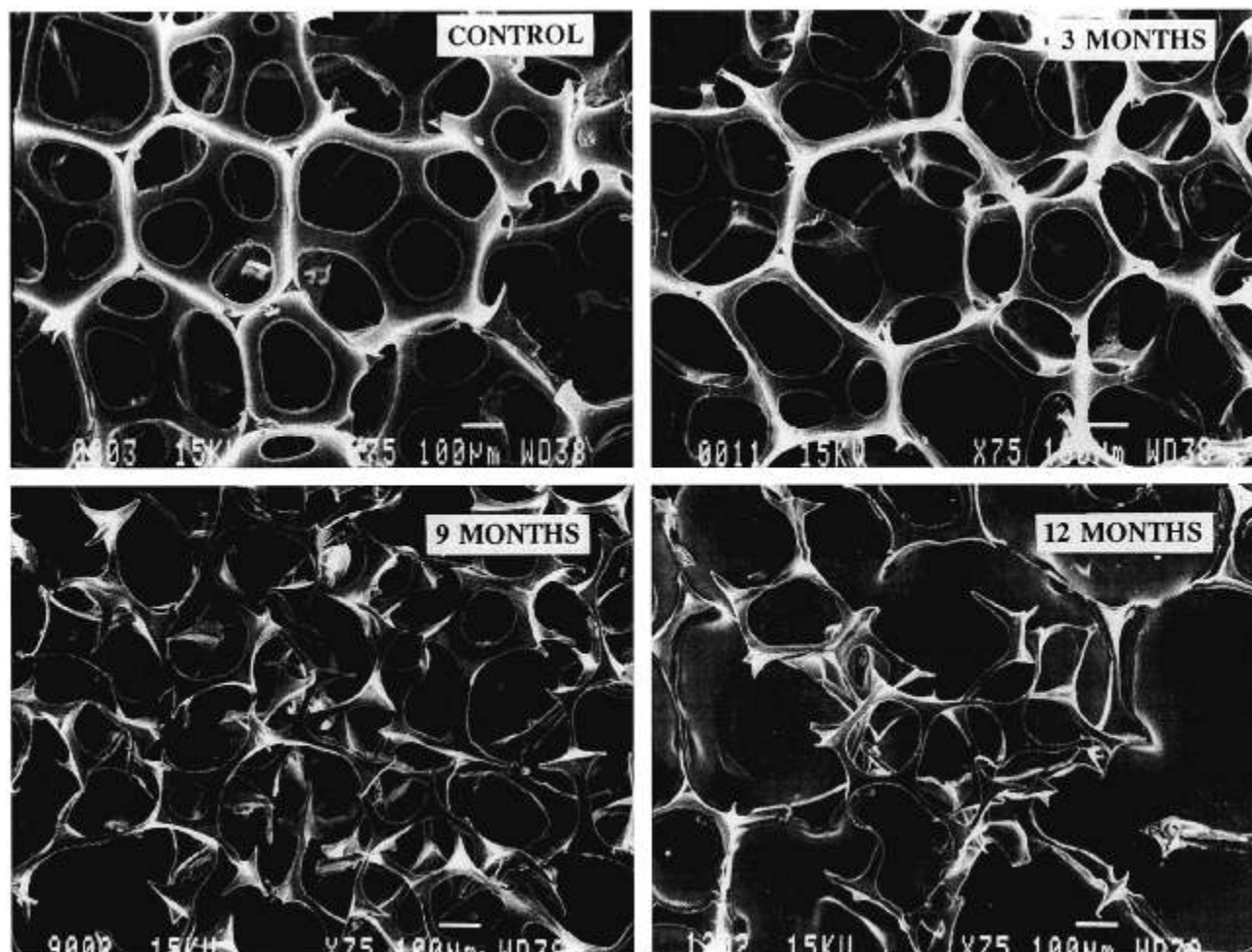


FIG. 4. Under higher magnification, there is progressive loss and collapse of the structural integrity of the open cell configuration with the foam struts thinned and fractured ($\times 75$).

maintaining eight implants and capsules were removed and prepared, and the animals were then sacrificed.

Analysis

Light Microscopy/Histology. The tissue-incorporated silicone-backed polyurethane foam implants that had been assigned the number 1 were fixed in formalin, embedded in paraffin, sectioned, stained with Giemsa, and examined by means of the light microscope. The analysis involved subjectively documenting the presence of a foreign-body inflammatory response as well as documenting any evidence of foam degradation.

Scanning Electron Microscopy. After the tissue capsules were digested enzymatically, the recovered polyurethane foam implants assigned the number 2 were prepared for scanning electron microscopic analysis as described in a previous publication.⁴ Images of the implants were pho-

tographed at magnifications of $12\times$, $75\times$, and $250\times$.

These photographs were used to record qualitative structural changes of the polyurethane struts and any distortion of the open cell network. Each $75\times$ photograph was analyzed by measuring the width of each of 20 struts at their thinnest portion. The average strut width was determined for each implant. Similar analysis was performed on control silicone-backed polyurethane foam.

Spectroscopy. Recovered polyurethane foam implants assigned the numbers 3 and 4 were stored in a dry container after digesting the tissue capsule enzymatically. These implants will undergo surface and bulk chemical analysis in the future. The results will be the subject of a separate report.

The animals' weights and the polyurethane strut widths are represented as a mean $\pm$ standard error of the mean. Analysis of variance was

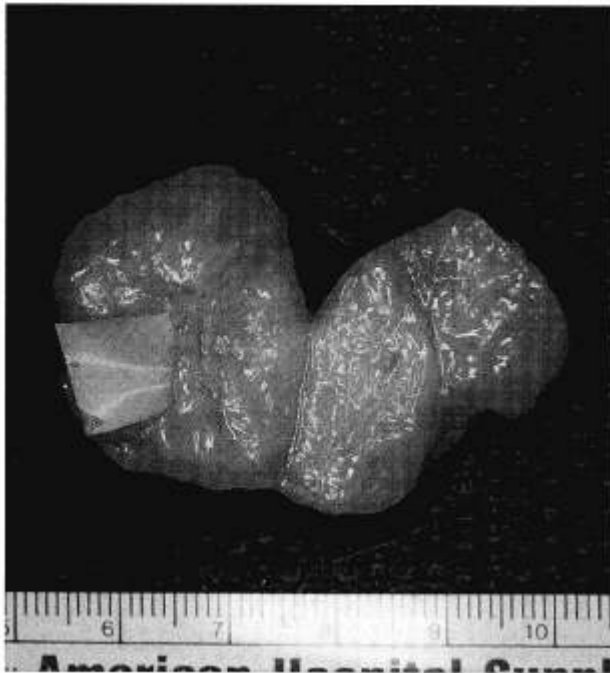


FIG. 5. Tumor sliced open to reveal the silicone elastomer backing without any foam. Histologically, this tumor was identified as a sarcoma.

used to test the significance of the effect of implantation time, animal weight, and foam manipulation on foam degradation.

Linear regression analysis also was used to estimate the rate of change in the size of the implant strut per month for the rat data as well as the human data from our previous study.⁴ This rate of change is equivalent to the slope of the linear function to which the data are fitted by the least-squares method. The precision and statistical significance of the slope were assessed by means of 95 percent confidence intervals.

RESULTS

The initial mean weight of the animals was 277.0 ± 6.3 gm. At the end of the experiment 1 year later, the animals had a mean weight of 487.3 ± 5.6 gm. A total of 27 of the 256 polyurethane implants were excluded from analysis. Eleven of these implants became exposed within the first month of implantation in five animals because the animals had either scratched or gnawed at the incision lines before healing had occurred. Two animals died after induction of anesthesia for implant removal at 9 months after implantation, so the remaining eight implants in each of these two animals could not be analyzed at the 12-month interval.

Histologic examination of the tissue capsule revealed a multinucleated giant cell foreign-

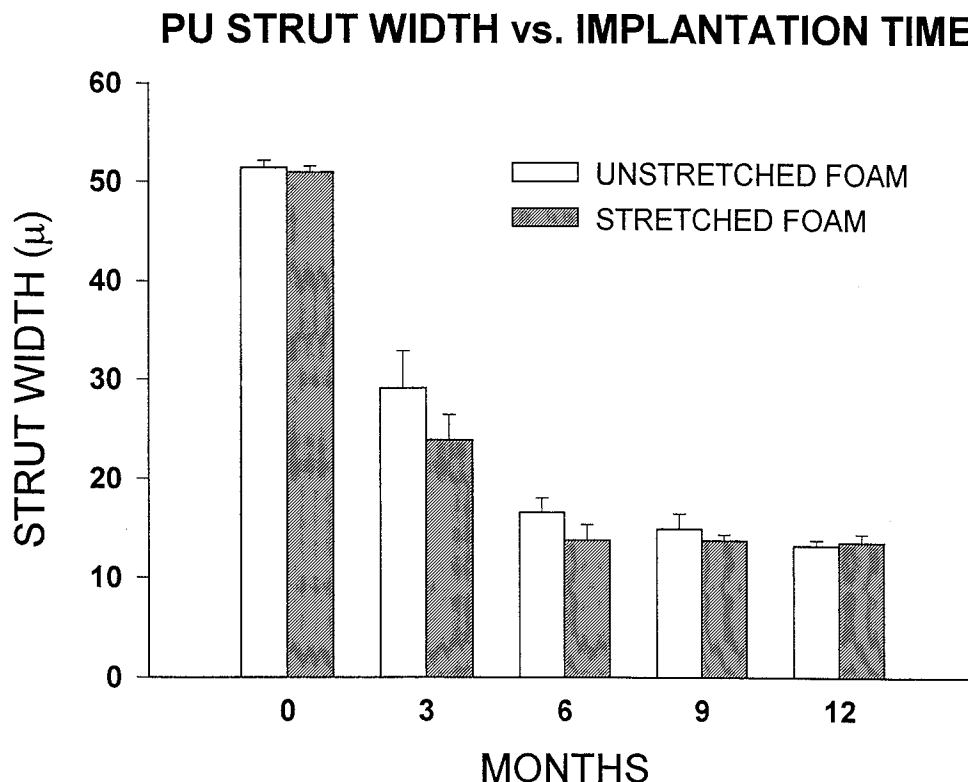


FIG. 6. The average polyurethane strut width in microns (mean $\pm$ SEM) versus the implantation time in months.

body inflammatory response. The typical triangular or hourglass shape of the foam cut in cross section was present in all capsules analyzed (Fig. 1). Subjectively, there was evidence of foam degradation. Many of the foam particles showed gradual blunting of the edges as the duration of implantation increased. There also was a subjective decrease in the amount of foam present but a relative increase in the amount of smaller foam particles. Turquoise-staining particles began to accumulate within many of the multinucleated giant cells (Fig. 2). These particles were seen only in tissue sections removed 6 to 12 months after implantation. Under the light microscope, subjectively there did not appear to be any difference in the structure of the foam or the tissue reaction between foam that had been manipulated preoperatively and foam that was unstretched.

Some foam was recovered from every implant whose tissue capsule was removed enzymatically. With the scanning electron microscope under low magnification, it was evident that a progressive loss of foam was occurring as the duration of implantation was increasing (Fig. 3). A rather rapid loss of the structural integrity of the open-pore structure was documented under higher magnification (Fig. 4). The foam

struts thinned out and fractured after 6 months of implantation. By 1 year, much of the intact foam had disappeared from the implants, but all the silicone elastomers remained intact.

One rat developed a tumor surrounding one of the implants approximately 9½ months after it was inserted. This tumor was left undisturbed until the animal was sacrificed at 12 months (Fig. 5). The tumor had grown around the implant, but no polyurethane foam was found on the elastomer backing. Histologically, the tumor was identified as a sarcoma, and no polyurethane particles were identified.

The duration of implantation was found to be a statistically significant variable with respect to the change in the foam strut width for both stretched and unstretched foam ($p < 0.0001$). Over the 1-year period, the foam strut width decreased by approximately 75 percent, from 51.4 ± 1.3 to $13.2 \pm 0.3 \mu\text{m}$. This change in strut width was similar for the stretched foam as well (Fig. 6). Strut width measurements from unimplanted control foam did not change over the 1-year period. The animals' weight was not a statistically significant variable to cause the change in foam strut width over time ($p < 0.057$).

Figures 7 and 8 depict the scatter plots of the

POLYURETHANE BIODEGRADATION - RAT

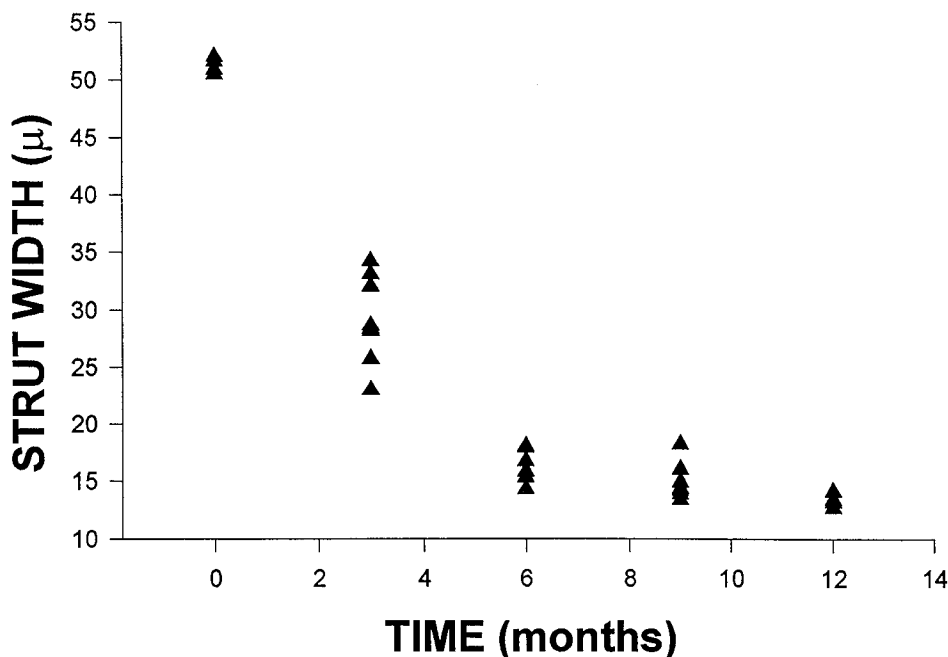


FIG. 7. Scatter plot of polyurethane foam strut width in microns versus implantation time in months.

POLYURETHANE BIODEGRADATION - HUMAN

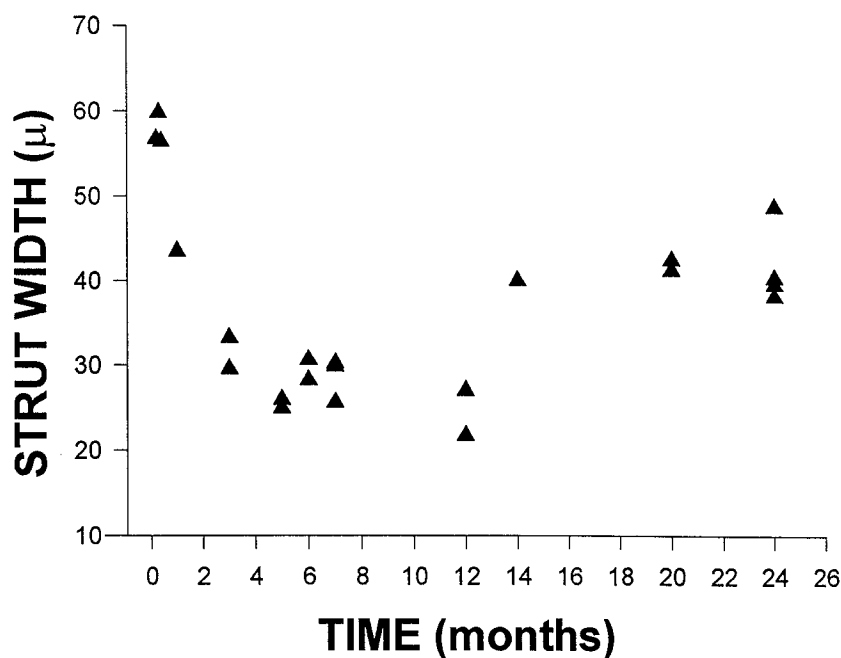


FIG. 8. Scatter plot of polyurethane foam strut width in microns versus implantation time in months from human data.

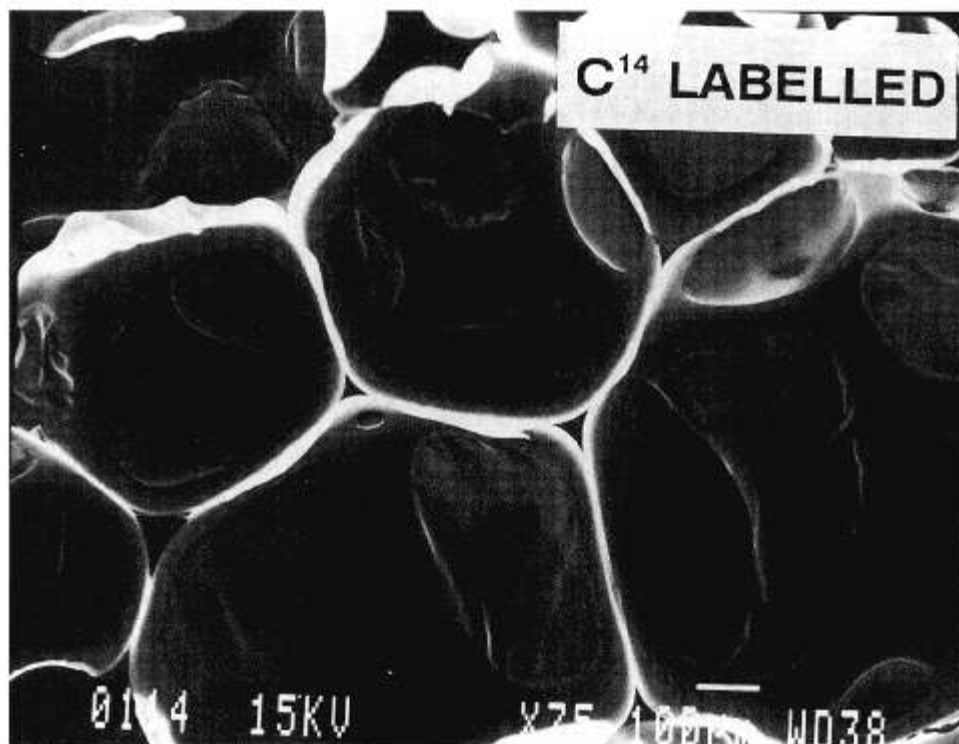


FIG. 9. Carbon-14-labeled polyurethane foam with a thin "skin" covering all the pores ($\times 75$).

implant strut width versus duration of implantation in months for the rat and human data, respectively. Table I shows the results of the linear regression analysis for the two data sets. The rat data provide evidence of a strong linear association between duration of implantation and strut width of the implant with a quadratic component that would account for the higher rate of change in the early time periods when compared with the later times. The linear regression analysis showed that the model explains a significant proportion of the variance in implant size ($r^2 = 0.74$, $p = 0.0001$) with a statistically significant slope ($b = -2.82$, $p = 0.0001$).

Contrary to these findings, the data from the previous human study failed to demonstrate a statistically significant linear relationship between duration of implantation and the implant strut width ($b = -0.12$, $p = 0.67$). The linear regression analysis failed to produce a model that could accurately predict strut width on the basis of duration, as indicated by the low coefficient of determination ($r^2 = 0.01$, $p = 0.67$).

DISCUSSION

Scientific literature examining the biocompatibility of the polyurethane covering of breast implants with human tissue is growing.^{1,3-9} In a previous study using histologic and scanning electron microscopic techniques, we were able to document evidence of structural changes of the polyurethane foam taken from human breast prostheses from a large number of patients.⁴ It was difficult to conclude definitively what factors affected the foam degradative process because of the large patient variation and the number of variables involved. Presently, it would be impossible for us to perform a prospective, randomized, controlled human study looking at the polyurethane degradation because this type of prosthesis is no longer manufactured for human implantation in North America. Although there have been various animal studies examining the tissue reaction to polyurethane,¹⁰⁻¹⁵ few studies have looked specifically at the structural changes in the foam

itself.^{16,17} An animal model that mimics the human experience would therefore be useful.

This prospective, controlled animal study uses the rat as our model. Our results indicate that similar structural changes are occurring in the rat foam implants as have been documented in human implants. Histologically, the same multinucleated giant cell foreign-body inflammatory reaction develops. This has been substantiated in other studies as well.^{13,17,18} Subtle structural changes occurring on the surface of the rat foam were similarly seen. What we did notice in the rats that has not been appreciable in human implant studies was the abundant amount of intracellular particles. This intracellular material appears to be made of smaller, degraded particles from the foam implants and presumably arises from an efficient phagocytic process by the multinucleated giant cells.¹⁹

The scanning electron microscopic analysis corroborates the light microscopic findings and was valuable in demonstrating the structural changes in the foam over time. Thinning of the foam struts and collapse of the open cell structure are constant findings during the degradative process of the foam. The rats appear to have an astounding ability to degrade the foreign material relatively quickly, as documented by the changes seen in the foam structure. Much of the foam had been broken down into smaller particles after only 1 year of implantation. The same degree of structural change is not usually present in human polyurethane-covered prostheses until 3 years after implantation. This difference in the rate of foam degradation is more impressive when one calculates the ratio of total foam implant surface area to body weight. Hypothetically, in a human patient weighing 60 kg with two polyurethane implants each having a base diameter of 10 cm, the patient's immune system would have to degrade approximately 1 cm² of foreign material per 200 gm of body weight. The rats in this study, however, had to contend with an average of 10 cm² of foreign material per 200 gm of body weight.

Although trauma or rough handling of polyurethane-covered prostheses during implantation might be considered a contributing factor in the premature breakdown of the foam documented clinically, this was not found to be a significant factor in this study. Excessive handling of the foam preoperatively had no effect, subjectively or objectively, on how the foam structure was degraded or on the rate of its degradation as measured by the strut widths.

TABLE I
Linear Regression Analysis

Model	Slope	95% Confidence Interval
Rat	-2.82	-2.23 to -3.40
Human	-0.12	+0.43 to -0.70

Concerning the rate of foam degradation in this rat study, we asked ourselves whether the results were comparable with our human study. Although we demonstrated a linear relationship between the foam strut width and duration of implantation in this animal model, the statistical analysis demonstrated that the biologic process of polyurethane foam degradation is different for the rat compared with the human when based on our previous data from explanted human prostheses. The monthly rate of change in foam strut width for the rat data was significantly different from the human data, making it unacceptable to extrapolate these results to the human.

In this study we could only document the relative rate of degradation of the polyurethane foam as measured by the change in the width of the foam struts. Ideally, we wanted to measure the absolute rate of degradation by labeling the foam with a radioisotope marker. This technique also would have helped us to understand how and where the foam was being eliminated. We were successful in obtaining carbon-14-labeled polyurethane foam (Polymedica Industries, Woburn, Mass.), which we examined under the scanning electron microscope. We found, unfortunately, that all the labeled foam had a "skin" that completely covered the pores (Fig. 9). Removing the skin would have required using, and ultimately contaminating, an industrial blast furnace or subjecting the foam to a strong acid that would have degraded the foam along with the skin. We therefore could not use the labeled foam for the experiment.

It is well recognized that rodents tend to develop sarcomas when exposed to solid-state implants. This phenomenon is referred to as the *Oppenheimer effect*.^{12,14,16,20-22} Hueper²³ has documented the development of carcinomas and sarcomas in relation to implanted polyurethane foam in rodents. Considering we had implanted a large number of silicone-backed foam implants, we expected to see a much higher number of tumors develop than we did. In fact, only one tumor developed, and this was associated with the silicone elastomer sheeting that had lost its porous foam surface. Although speculative, it is interesting to note that the porous surface of the foam may have had a preventative effect on sarcoma formation in our rat study. This finding is supported by studies from Brand,^{14,20} who noted almost a 30 percent tumor incidence in mice with uncoated silicone implants but no tumors associated with poly-

urethane-coated silicone implants. Unlike rodent studies, it must be stressed that, to date, there is no scientific evidence linking sarcomas or carcinomas with the implantation of any type of breast prosthesis, or other prostheses, in humans.²⁴⁻²⁷

CONCLUSION

In conclusion, in a controlled, randomized animal study, the rate of polyurethane foam degradation in the rat appears to be linear as measured by the foam strut width. This linear relationship could not be extrapolated to the human when based on previous data of human explanted polyurethane-covered breast prostheses. Subjectively, this animal model demonstrated a much faster degradation of the foam than previously observed in human studies. The rat, however, is a good animal model to mimic the tissue reaction and the polyurethane foam structural changes documented in human studies but is a poor model to predict rates of biodegradation in humans.

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